



# Revisiting the construct and assessment of industrial relevance in software engineering research

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## ABSTRACT

Industrial relevance is essential for an applied research area like software engineering. However, it is unclear how to achieve industrial relevance and how we communicate and assess it. We propose a reasoning framework to support the design, reporting, and assessment of research for industrial relevance.

## 1 INTRODUCTION

Industrial or practical relevance is an essential aim for software engineering research. To improve industrial relevance, the literature suggests using appropriate research approaches, selecting relevant research problems and closer collaboration with industry [7]. To support those tasks, specific technology transfer models [8], interactive research approaches [14, 15], and technology adoption theories [4] have been proposed. These suggestions approach the problem from a research planning point of view. However, they do not support other researchers or practitioners in assessing the industrial or practical relevance of published research results.

Some approaches to assessing research relevance have been proposed. Ivarsson and Gorschek's [10] assessment model for rigor and (industrial) relevance, for example, considers whether certain context information was reported or whether an intervention was used by practitioners. However, knowing the context and that practitioners used the intervention does not necessarily imply that the intervention is relevant to the context or the practitioners. Engström et al. [6] suggest that relevance should be assessed either in general terms of the prevalence and severity of the addressed problem or for a specific situation in terms of the applicability and potential effect of the proposed intervention from the practitioners' point of view. In both cases, support for this assessment is needed.

Existing definitions of relevance show that there is no commonly accepted view of the meaning of relevance (e.g., [2, 3, 7, 13, 16]). While most definitions point out context as a central attribute, it can be noted that there are varying views on the attributes that constitute relevance. Some definitions focus on applicability, however,

without describing how applicability can be described or assessed.

Here, we reflect on characterizations of relevance as a basis to work toward assessing it. We make the following contributions:

- (1) Provide and compare definitions and attributes of relevance in key publications.
- (2) Provide an initial proposal for achieving, reporting, and assessing industrial relevance in software engineering.
- (3) Reflect on the proposal as a basis for reasoning on relevance.

## 2 RESEARCH RELEVANCE – DEFINITIONS AND ATTRIBUTES

Industrial or practical relevance is an important quality factor for any applied or interdisciplinary field. Belcher et al. [3] identify relevance as one of the four main principles of research quality in trans-disciplinary research. They define relevance as “[t]he importance, significance, and usefulness of the research problem, objectives, processes, and findings to the problem context” and suggest a rubric for seven aspects of relevance. Six of these seven aspects deal with the problem context. Similarly, relevance is one of three key quality dimensions in Hevner's three-cycle view of design science research [9], next to rigor and novelty.

Relevance is also a key concept in information retrieval and has been studied since the late 50s. Saracevic [16, pp. 26–27] describes five kinds (manifestations) of relevance that are further described by ten aspects: Relation, intention, context (overall, internal, and external), inference, selection, interaction, measurement, and truth. Notably, he claims that relevance cannot be considered without a context, that relevance may change, and that relevance can be measured, i.e., assessed: “Relevance involves a graduated assessment of the effectiveness or degree of maximization of a given relation, such as assessment of some information sought, for an intention geared toward a context.”

Garousi et al. [7] investigated practical relevance for software engineering research and proposed three dimensions of practical relevance: A focus on real-world software engineering problems, applicability (implementability), and actionable implications for practitioners. Like the two previous proposals, Garousi et al. are also concerned with the problem context. In addition to context, Ali et al. [2] considered the expected effect and characteristics of techniques in their search for relevant regression testing research.

Molléri et al. [13] looked more broadly into research quality in empirical software engineering. They propose a model for research quality based on the model for the quality of research practice proposed by Mårtensson et al. [12]. Both quality models group

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their quality attributes into four categories: credibility, contribution, communication, and conformance. In both models, relevance is a subcategory of contribution. In Molléri et al., relevance is explicitly related to software engineering practice.

Table 1 provides an overview of the definitions used within software engineering and related fields.

Additionally, we extracted attributes of research relevance across different references. Table 2 provides a list of attributes and shows the attributes shared across the studies in Table 1 (Belcher et al. [3], Saracevic [16], Garousi et al. [7], Ali et al. [2], Ivarsson and Gorschek [10], and Molléri et al. [13]).

Table 2 shows that relevance covers a wide range of attributes. When designing our solution, we later contrast our solution proposal’s coverage of attributes with these attributes.

### 3 ASSESSING RELEVANCE

The critical appraisal of the quality of published research has a long tradition in the medical and health sciences. A systematic review from 2004 identified 121 critical appraisal tools and concluded that 87% of them were specific to a research design and that there is no “widely accepted generic tool that can be applied equally well across study types” [11].

A general challenge with appraisal rubrics is that they may be used mechanistically by “ticking off” the existence of descriptions of required aspects. That makes sense since existence is a necessary condition for relevance assessment. Existence of description is not sufficient, though. In addition, the reported aspects must be meaningful to the readers or assessors, for example, aligning with their specific intentions, goals, and contextual needs.

Furthermore, we should distinguish between papers enabling “relevance assessment” and papers actually reporting relevant work. As relevance is a subjective matter, we should maybe not assess relevance in absolute terms at all.

Sutherland et al. [17] distinguish four levels of impact and relevance assessment (pure research, applied research, dissemination/development, and societal benefits) with associated measures for impact and relevance that are progressively more difficult to obtain, the more the science progresses into practice.

From the design science perspective, relevance would be assessed differently by researchers and practitioners. While the research community is interested in relevance in general terms, i.e., in terms of the prevalence and severity of the addressed problem and the potential of the proposed solution, practitioners would most likely look at relevance from the perspective of their specific situation, i.e., in terms of the applicability and potential effect of the proposed intervention from the practitioners’ point of view [6]. For both cases, support for making these assessments needs to be reported.

### 4 A REASONING FRAMEWORK FOR ASSESSING RELEVANCE (RFR)

Industrially relevant research contributes in one or more of the following ways, as described in the design science framework [6]: (a) in the theoretical problem space (as an understanding of a problem or an improved understanding of a previously researched problem), (b) in the theoretical solution space (as a design of a new solution or improvement to an existing solution), (c) in the practical solution

space (as an experience report of implementing a solution design to a concrete case) detailing an implementation or an instantiation of a solution, or (d) in the practical problem domain as an empirical in-context validation of a solution.

We propose a framework to reason about the industrial relevance of a research contribution (in any of the four spaces listed above) from a design, reporting, or assessment perspective. It draws inspiration from Zachman’s framework for enterprise architectures [18] and Dybå’s omnibus approach to describe the context for empirical research [1, 5]. The resulting framework, as shown in Table 3, has six aspects (questions) that are considered from three perspectives: design, reporting, and assessment. The six aspects are briefly described below:

**What** refers to the proposed solution design or the refined/improved understanding of the problem.

**How** refers to the procedures followed and resources required to use the solution or the refined/improved problem characterization.

**Where** deals with the location and settings.

**Who** refers to the roles, responsibilities, and stakeholders.

**When** refers to timing, e.g., in terms of organization, product or software development life-cycle, or the span of the research.

**Why** refers to reasons for the choices made in the previous concerns/questions.

We now briefly describe how the decision regarding these six aspects may be informed, reported, and assessed from the goal of conducting and reporting industrially relevant research.

**Designing:** For any investigation in the problem or solution space, the goal of conducting industrially relevant research and the problem being investigated or addressed (**what**) will inform our choices (**why**) of the settings (**where**), the time (**when**), and the people (**who**) to involve in the investigation.

**Reporting:** Primarily, provide sufficient details to answer the following: **What** is the problem? **Why** is it a problem (building on **how**, i.e., the characterization of the problem, **when**, for **whom**, and **where** it is a problem)? Similarly, for the solution space, one could describe the following: **What** is the solution? **Why** is it a solution (building on **how**, i.e., solution specification and resources required, **when**, i.e., applicable time/phase, for **whom**, i.e., the problem owners, and **where**, i.e., the settings)?

**Assessment:** A researcher or a practitioner trying to assess the industrial relevance of research has to describe the problem for which they are looking for a better understanding or a potential solution. They should use the same six aspects to describe their problem before attempting to assess the relevance of the research work.

Considering this problem statement, an assessor may answer the following question relying on the reporting of the six aspects:

Does the solution meet our needs? That is, is the solution (**what**) likely to work for us, considering the target user (**for whom**), the resources required (**how**), the applicable time/phase (**when**), the settings (**where**), and the strength of evidence (**why**) reported by research. Similarly, one could answer, does the problem investigation provide/improve our understanding of the problem?

Table 1: Definitions of relevance.

| Reference (field): Definition                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Belcher et al. [3] ( <b>Transdisciplinary</b> ): Relevance is the importance, significance, and usefulness of the research problem, objectives, processes, and findings to the problem context.                                                                                                                                                                                                                                                                          |
| Saracevic [16] ( <b>Information retrieval</b> ): The author describes five “manifestations” of relevance, of which three are relevant here: Topical or subject relevance (the relation between the searched topic and the found information), pertinence or cognitive relevance (the relationship between a user’s knowledge and the found information), and situational relevance (the usefulness of the found information for the task, situation or problem at hand). |
| Garousi et al. [7] ( <b>Software Engineering</b> ): Industrial (practical) relevance of a research paper or a research project is the degree of its relevance and potential value for software engineering organizations and industry.                                                                                                                                                                                                                                   |
| Ali et al. [2] ( <b>Software Engineering</b> ): Industrial (practical) relevance of a proposed intervention is defined as a combination of desired (or measured) effects, addressed context factors, and the characteristics of the technique that are critical for applicability.                                                                                                                                                                                       |
| Ivarsson and Gorschek [10] ( <b>Software Engineering</b> ): Industrial relevance concerns the evaluation’s value for practitioners seeking to adopt technologies.                                                                                                                                                                                                                                                                                                        |
| Molléri et al. [13] ( <b>Software Engineering</b> ): The research addresses a new problem by means of original theory, method and/or context.                                                                                                                                                                                                                                                                                                                            |

Table 2: Consolidated attributes of research relevance across different references.

| Attribute                                            | [3] | [16] | [7] | [2] | [10] | [13] |
|------------------------------------------------------|-----|------|-----|-----|------|------|
| Context (e.g., internal, external, socio-ecological) | ✓   | ✓    |     | ✓   | ✓    |      |
| Socially relevant research problem                   | ✓   |      |     |     |      |      |
| Engagement with problem context                      | ✓   |      |     | ✓   |      |      |
| Explicit theory of change                            | ✓   |      |     |     |      |      |
| Relevant research objectives and design              | ✓   |      |     | ✓   | ✓    |      |
| Appropriate project implementation                   | ✓   |      |     |     |      |      |
| Effective communication                              | ✓   |      |     |     |      |      |
| Relation and intention                               |     | ✓    |     |     |      |      |
| Inference, selection, and interaction                |     | ✓    |     |     |      |      |
| Measurement and truth                                |     | ✓    |     |     | ✓    |      |
| Focusing on real-world problems (SE or other)        |     |      | ✓   | ✓   | ✓    | ✓    |
| Applicable and actionable                            |     |      | ✓   | ✓   |      | ✓    |
| Current and interesting                              |     |      |     |     |      | ✓    |
| Readable (documentation clarity)                     |     |      |     |     | ✓    | ✓    |

## 5 A PERSPECTIVE FROM THE INDUSTRY

We discussed relevance with a practitioner with 15 years of experience for preliminary input from the practitioners’ perspective. We summarize the conversation below and highlight the key aspects suggested by the practitioner in boldface text. In the future, we plan a more thorough investigation involving more practitioners.

According to the practitioner, they judge relevance based on their **role & responsibilities** and **current needs** at the company, and then to evaluate the relevance, they consider the **scale** at which a technique has been validated. However, they consider small-scale validations as relevant depending on the **novelty** of the solution. For disruptive innovations, “major breakthroughs”, they look for blogs and other sources instead of research papers. **Personal interest** in a topic “is a small factor too”, it also influences what one considers as relevant.

Regarding “role and responsibilities”, they elaborated that as their role is that of an architect and test quality manager, i.e., mainly working with development and testing, they are mainly interested in research on these topics. “For example, I am generally not interested in requirements engineering research”. Regarding the “current needs”, they emphasized that at companies, often the prioritization of open challenges changes, which means that a topic and the research on the topic that was relevant “six months” ago may not be considered relevant anymore.

## 6 DISCUSSION

### 6.1 Comparison with related work

Compared with the *related work* (definitions and attributes of relevance in the literature), our RFR framework considers a broader spectrum of factors – encompassing ‘What’, ‘How’, ‘Where’, ‘Who’, ‘When’, and ‘Why’. This holistic view allows for more detailed and structured reasoning on research contributions in the field. For example, Belcher et al.’s [3] focus on the problem context is integrated in our framework. At the same time, we extended the attributes of relevant research to include specific questions related to the design, reporting, and assessment of research. Overall, the RFR framework’s added value lies in its ability to systematically assess the industrial relevance of research across multiple dimensions. Table 4 maps the attributes identified in the related work to our framework. The table shows that our framework has more comprehensive coverage of relevance dimensions than the software engineering references (cf. [8, 13]). Additionally, it distinguishes the perspectives of designing, reporting, and assessing. Thereby, we take into consideration that the reflections on relevance differ depending on whether the research is in the design, write-up, or assessment stage.

### 6.2 Contrasting the framework with the practitioner’s perspective

Next, we contrast the framework and its structure with the views raised by the practitioner, discussing the dimensions of roles and responsibilities, current needs of the company, scale of validation, novelty of the solution, and personal interest.

The RFR considers the ‘Who’ aspect, focusing on *roles, responsibilities, and stakeholders*. This aligns with the practitioner’s emphasis on the relevance of research based on their specific role within the organization (e.g., architect, test quality manager). The RFR may not, however, fully address how different roles might prioritize different aspects of research.

The ‘What’ and ‘Why’ aspects of the RFR could encompass the *current needs and motivations of a company*. The RFR might not explicitly mention the change of industry needs over time, as highlighted by the practitioner’s point about the shifting relevance of topics.

The ‘How’ and ‘Where’ aspects might indirectly cover the *scale of validation*, as they deal with the implementation and setting of the solution. The scale at which a technique has been validated could be inferred from these details.

Novelty can be seen as part of the ‘What’ and ‘Why’ aspects,

**Table 3: A reasoning framework for industrial relevance (RFR) inspired by Zachman [18] and Dybå [5].**

|                    | What (solution/problem) – How (solution/problem specification and required resources) – Where (settings) – Who (problem owner, stakeholders) – When (phase) – Why (motivation)                                                                                                                                                                                                                                                                                     |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Perspective</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Designing</b>   | <b>Solution design considerations</b> when aiming for industrial relevance – i.e., when validating a solution or investigating a problem, decide where it should be investigated, who should be involved in the investigation, when should the investigation take place?                                                                                                                                                                                           |
| <b>Reporting</b>   | <b>Evidence:</b> What did we find works explaining why taken into consideration how it was implemented, what resources were required, where it was implemented, for whom and when? What did we find is a problem by describing how it becomes a problem, where, for whom, and when?<br><b>Theoretical/statistical generalizability:</b> What will likely work, how, where, for whom, when, and why? What is likely to be a problem, how, where, for whom, and why? |
| <b>Assessing</b>   | <b>Argumentative assessment:</b> Will it work to solve our problem, explaining why considering how (what will it take), where (for our settings), for whom (the problem owner), when (in the given phase)?                                                                                                                                                                                                                                                         |

**Table 4: Mapping of research relevance attributes to the RFR framework.**

| Attribute                                     | Reflection in RFR framework                                           |
|-----------------------------------------------|-----------------------------------------------------------------------|
| Context                                       | Addressed in 'Where' (settings).                                      |
| Socially relevant research problem            | Partially addressed in 'What' (solution/problem) aspect.              |
| Engagement with problem context               | Addressed in 'Where' and 'Who' (stakeholders) aspects.                |
| Explicit theory of change                     | Not directly addressed.                                               |
| Relevant research objectives and design       | Addressed in 'What' and 'How' (procedure/resources) aspects.          |
| Appropriate project implementation            | Addressed in 'How' aspect.                                            |
| Effective communication                       | Not directly addressed.                                               |
| Relation and intention                        | Addressed in 'What' and 'Why' aspects.                                |
| Inference, selection, and interaction         | Addressed in 'How' and 'Why' aspects.                                 |
| Measurement and truth                         | Partially addressed in 'How' (e.g., resource requirements) and 'Why'. |
| Focusing on real-world problems (SE or other) | Core to 'What' and 'Who' aspects.                                     |
| Applicable and actionable                     | Addressed in 'How' and 'When' (timing/phase) aspects.                 |
| Current and interesting                       | Partially addressed in 'What' and 'Why' aspects.                      |
| Readable (documentation clarity)              | Not directly addressed.                                               |

especially if the solution or problem understanding is *novel and innovative*. The framework encourages a comprehensive understanding of the problem and solution, which could include evaluating their novelty. There might be a need for more explicit guidance within the RFR on how to assess or report the novelty aspect.

*Personal interest* is not explicitly covered in the RFR. The framework primarily focuses on a structured, objective assessment rather than subjective factors like personal interest. The lack of emphasis on personal interest might limit the framework's ability to capture the practitioner's engagement with the research, which can be an important driver of perceived relevance.

Overall, the RFR aligns with some of the dimensions raised by the practitioner, particularly in addressing roles, responsibilities, and the nature of problems and solutions. However, it could be further refined to explicitly include the dynamics of industry needs, the scale of validation, the novelty of solutions, and personal interest.

## 7 CONCLUSION

This paper discusses the essential but difficult-to-define construct of industrial relevance. We present a reasoning framework for assessing relevance (RFR) that provides directions for future research. By comparing the framework with related work and the view of one expert, future directions for developing the framework further were highlighted.

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